Microsoft

AZ-040

Automate Administration with PowerShell

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Learning Path 9: Manage cloud resources by using Windows PowerShell

Learning objectives

- Review the Azure PowerShell module
- Review the features and tools for Azure Cloud Shell
- Manage Azure resources with Windows PowerShell

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Agenda

LP 1 Get started with Windows PowerShell

LP 2 Maintain system administration tasks in Windows PowerShell

LP 3 Work with the Windows PowerShell pipeline

LP 4 Work with PowerShell providers and PowerShell drives in Windows PowerShell

LP 5 Querying management information by using CIM and WMI

LP 6 Use variables, arrays, and hash tables in Windows PowerShell scripts

LP 7 Create and modify scripts by using Windows PowerShell

LP 8 Administer remote computers by using Windows PowerShell

LP 9 Manage cloud resources by using Windows PowerShell Azure

LP 10 Manage Microsoft 365 services by using Windows PowerShell M365

LP 11 Create and manage background jobs and scheduled jobs in Windows PowerShell

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Overview

Besides managing local resources, you can also use Windows PowerShell to manage cloud resources, such as Azure platform or Microsoft 365. To manage cloud-based resources, you need to install additional modules for Windows PowerShell. In this Learning Path, you'll learn how to install modules necessary for cloud services management to Windows PowerShell, and you'll also learn how to use PowerShell commands to perform some simple administrative tasks on cloud resources such as Azure virtual machines (VMs), Azure storage accounts, and Azure subscriptions. You'll also learn how to use Azure Cloud Shell environment to perform PowerShell based on Bash based administration directly from Azure portal.

Modules:

- Review the Azure PowerShell module
- Review the features and tools for Azure Cloud Shell
- Manage Azure resources with Windows PowerShell

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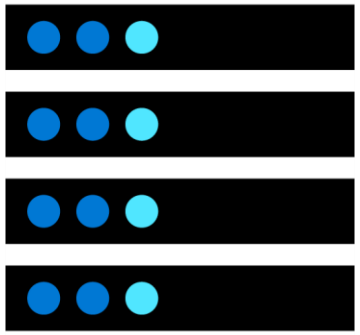
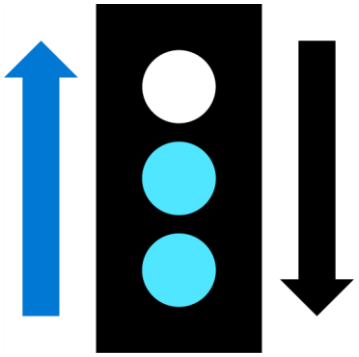
What is cloud computing?

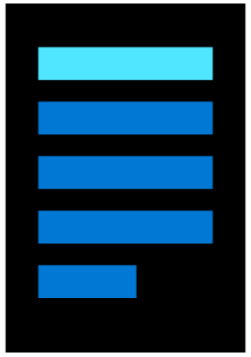
Cloud computing is the delivery of computing services over the internet, enabling faster innovation, flexible resources, and economies of scale.



Compute



Networking



Storage

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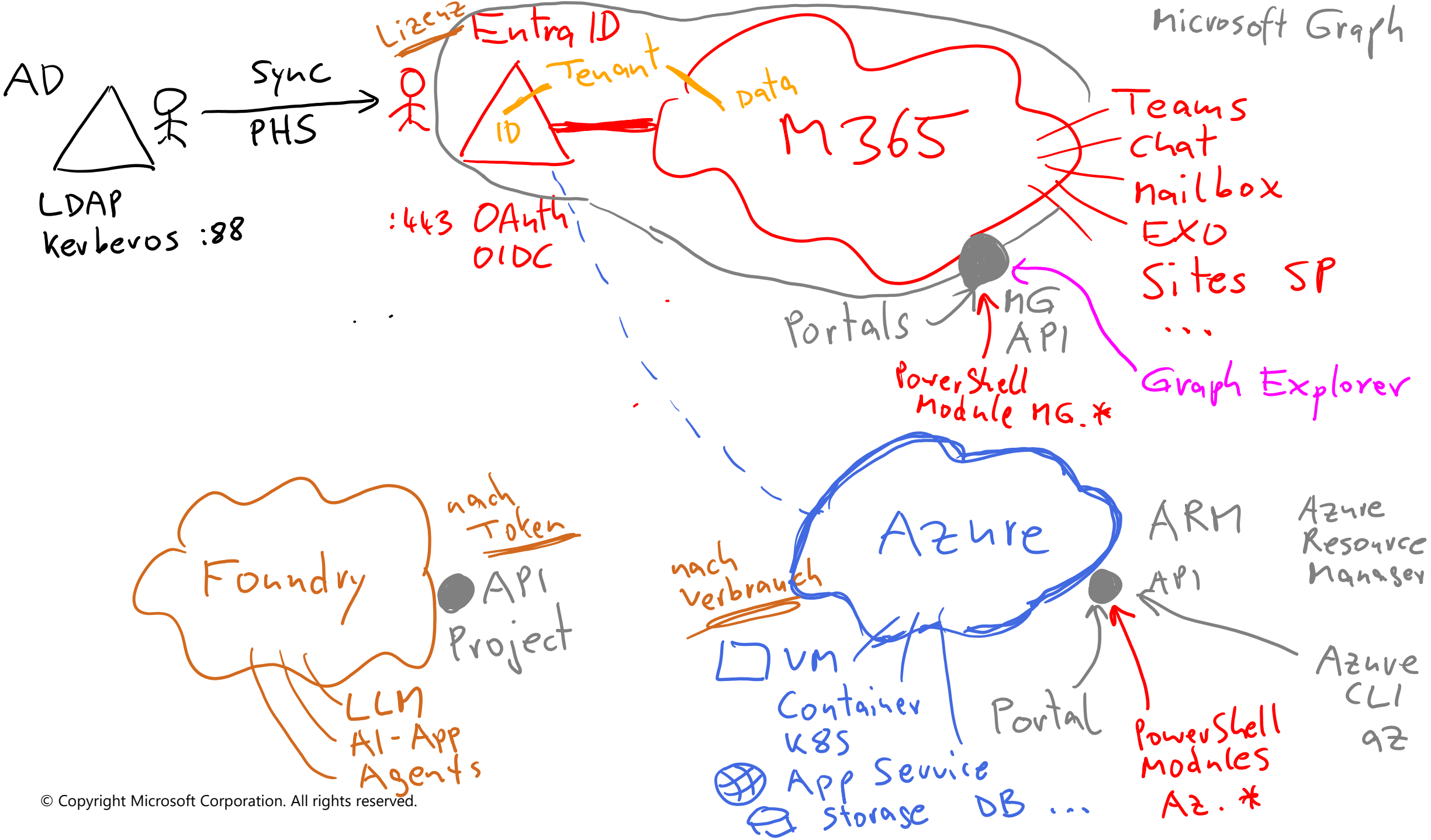

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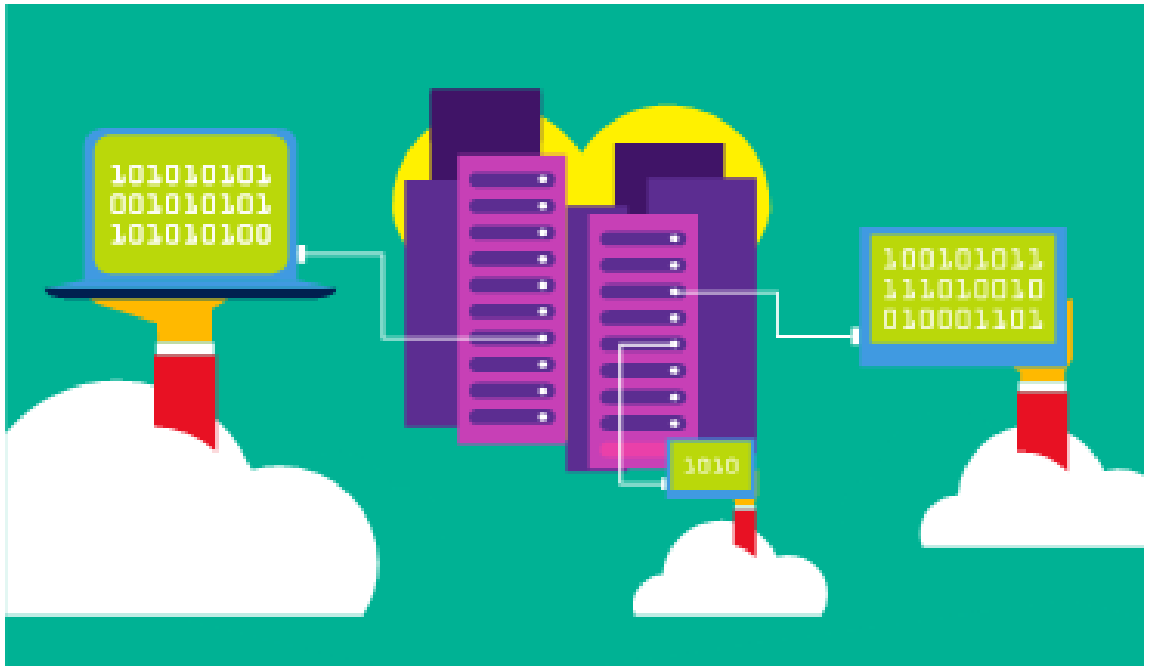
Public cloud

- Owned by cloud services or hosting provider.
- Provides resources and services to multiple organizations and users.
- Accessed via secure network connection (typically over the internet).



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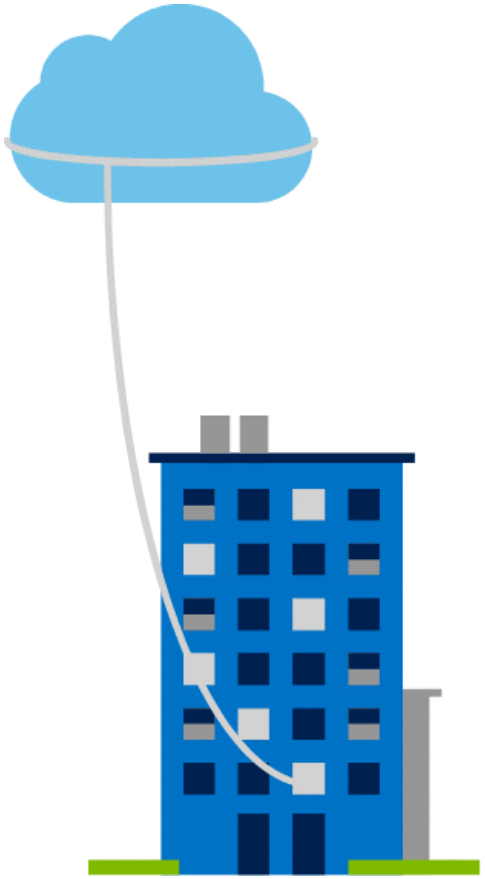
Private cloud

- Organizations create a cloud environment in their datacenter.
- Organizations are responsible for operating the services they provide.
- Does not provide access to users outside of the organization.



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Hybrid cloud

Combines **public** and **private** clouds to allow applications to run in the most appropriate location.

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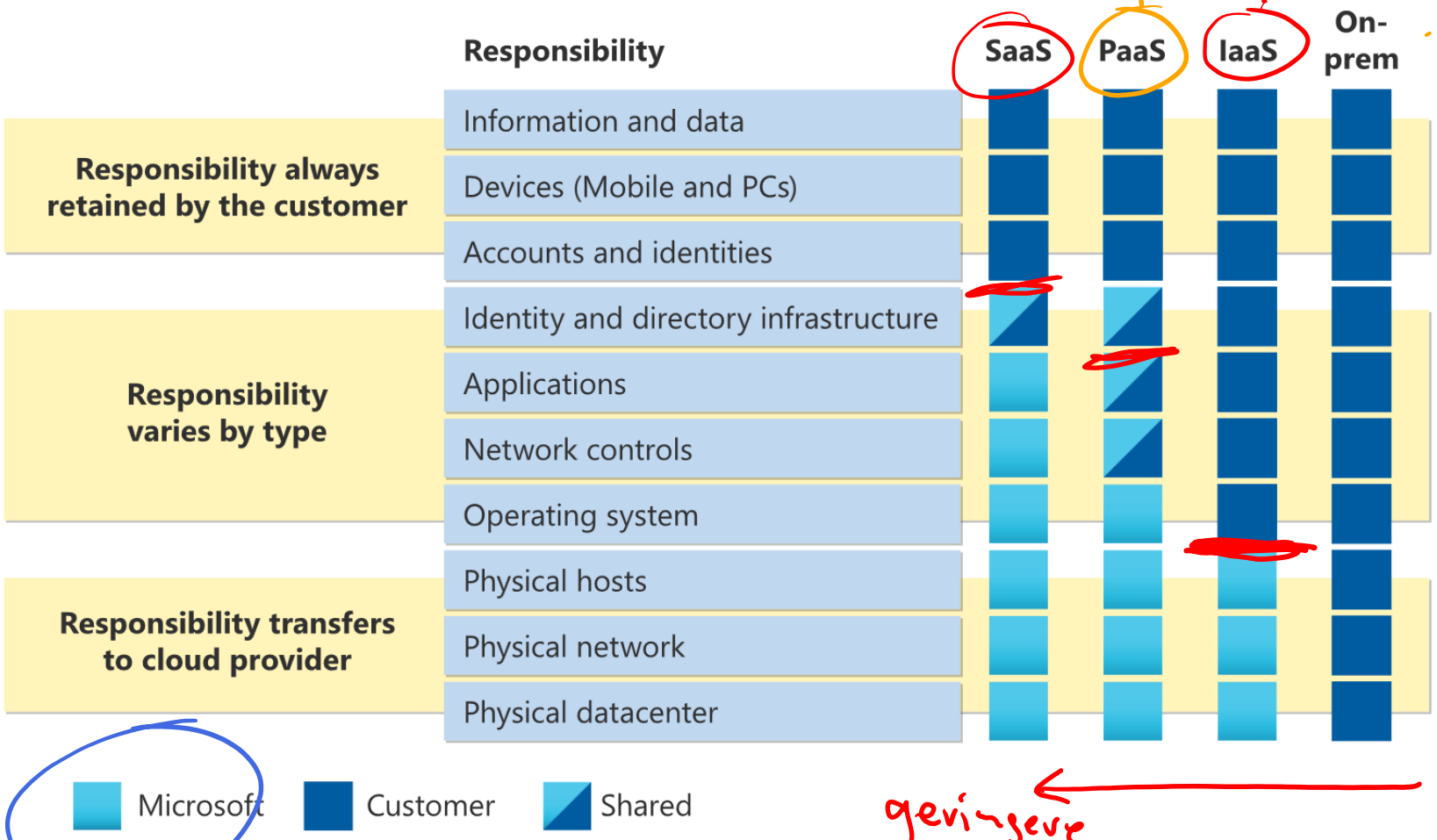
Shared responsibility model

.Net Code
web site
App Service
VM



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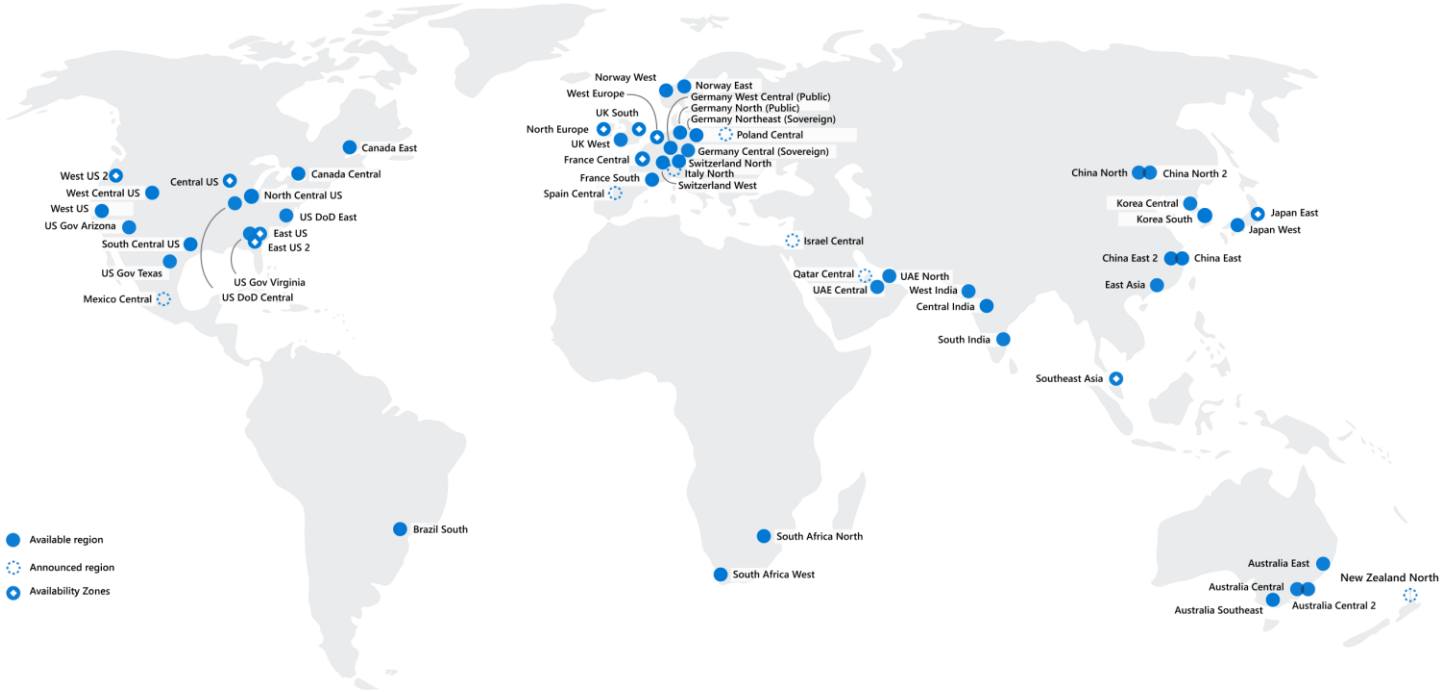
Regions

Azure offers more global regions than any other cloud provider with 60-plus regions representing over 140 countries



- Regions are made up of one or more datacenters in close proximity.
- They provide flexibility and scale to reduce customer latency.
- Regions preserve data residency with a comprehensive compliance offering.

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Azure resources

Azure **resources** are components like storage, virtual machines, and networks that are available to build cloud solutions.

Virtual machines

Storage accounts

Virtual networks

App services

SQL databases

Functions

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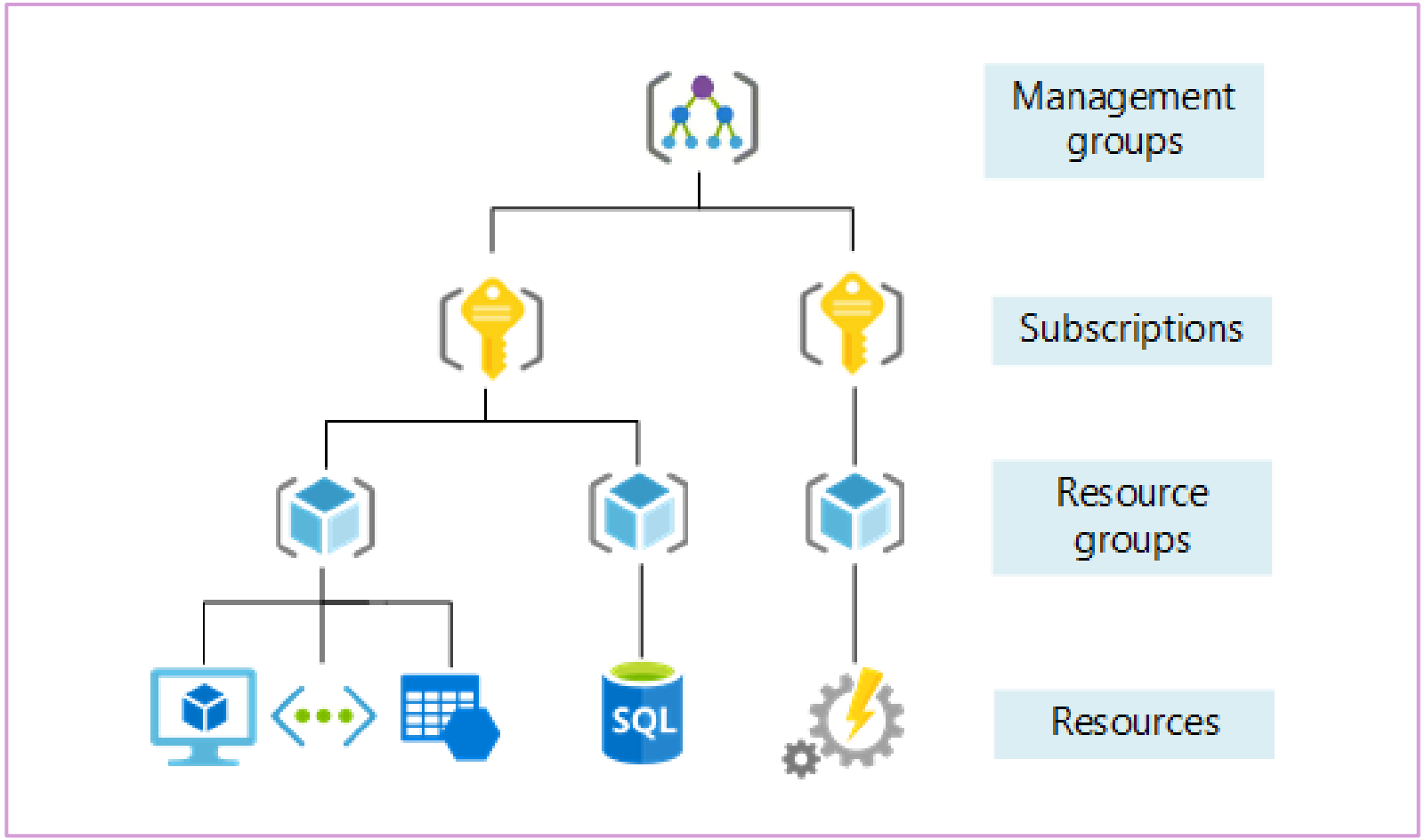
Management groups

- Management groups can include multiple Azure subscriptions.
- Subscriptions inherit conditions applied to the management group.
- 10,000 management groups can be supported in a single directory.
- A management group tree can support up to six levels of depth.



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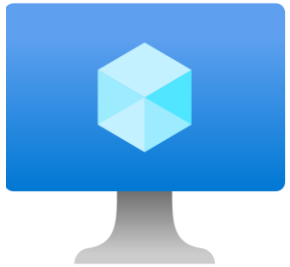
Azure compute services

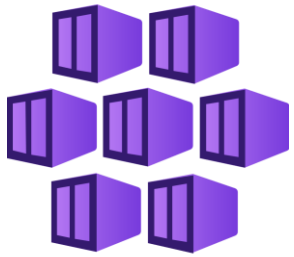
Azure **compute** is an on-demand service that provides computing resources such as disks, processors, memory, networking, and operating systems.



Virtual Machines

App Services

Container Instances



Azure Kubernetes Services (AKS)

Azure Virtual Desktop

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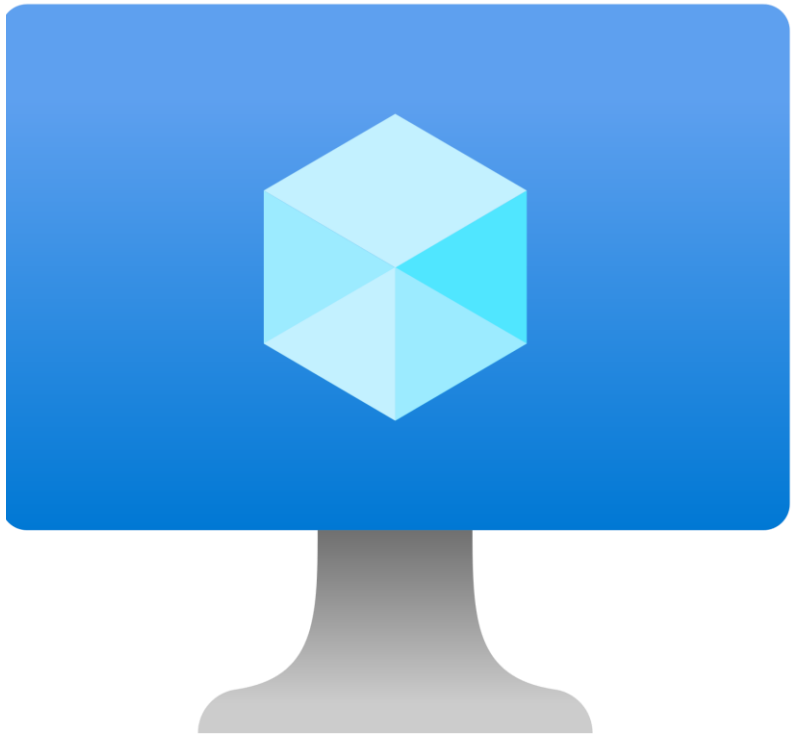
Azure virtual machines

Azure **virtual machines (VMs)** are software emulations of physical computers.

- Includes virtual processor, memory, storage, and networking.
- IaaS offering that provides total control and customization.



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Find-Module Az | Install-Module

Azure Resource Manager

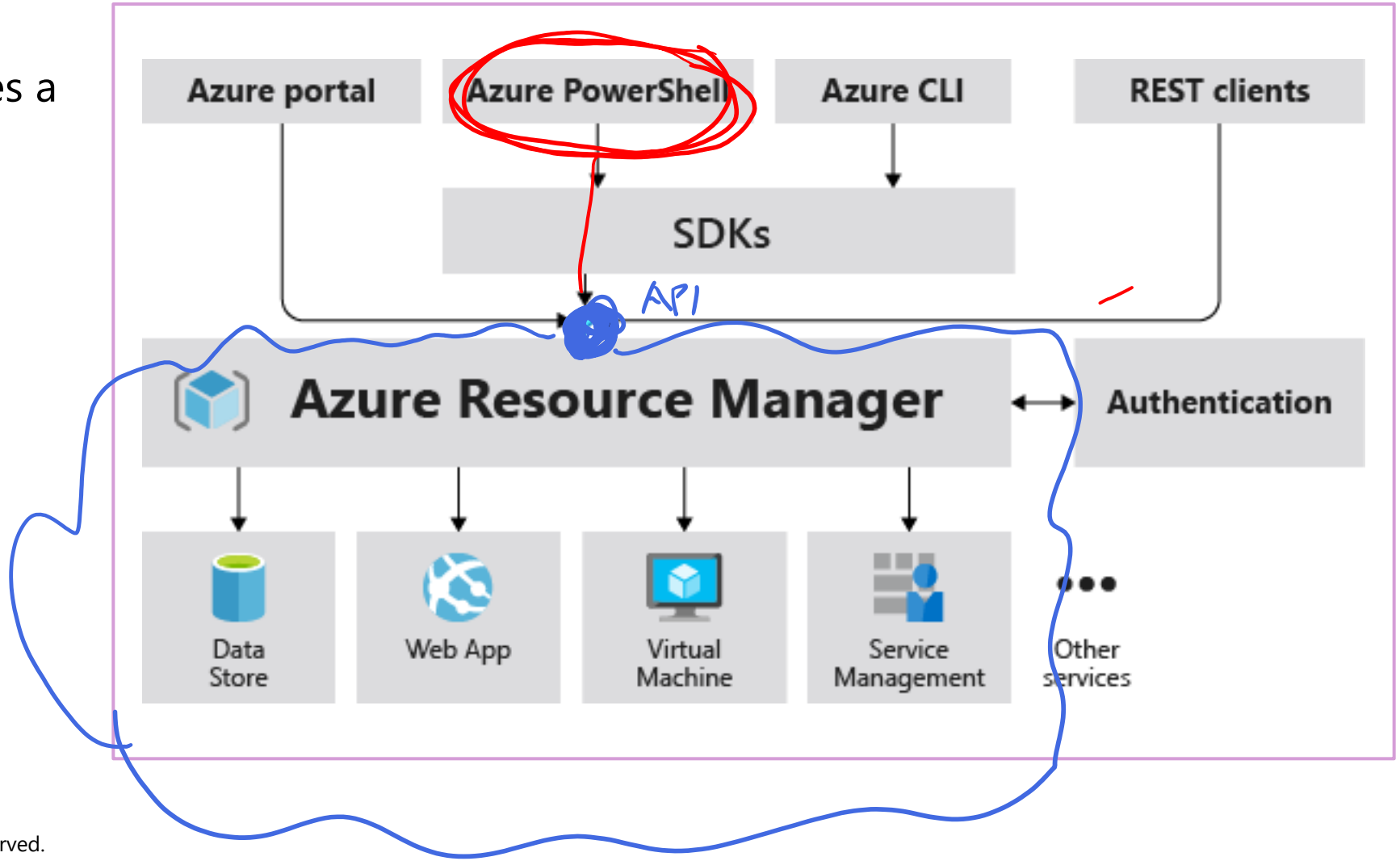
Module Az

Az.Compute

The **Azure Resource Manager (ARM)** provides a management layer that enables you to create, update, and delete resources in your Azure subscription.

Get-AzVm

Get-Vm



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Pause bis 15:15

Azure Portal

Terminal
Get-AzVM

Azure "Cloud Shell"
Docker Container

Az
Linux

Window Terminal

Terminal
Get-AzVM

VS Code

Terminal

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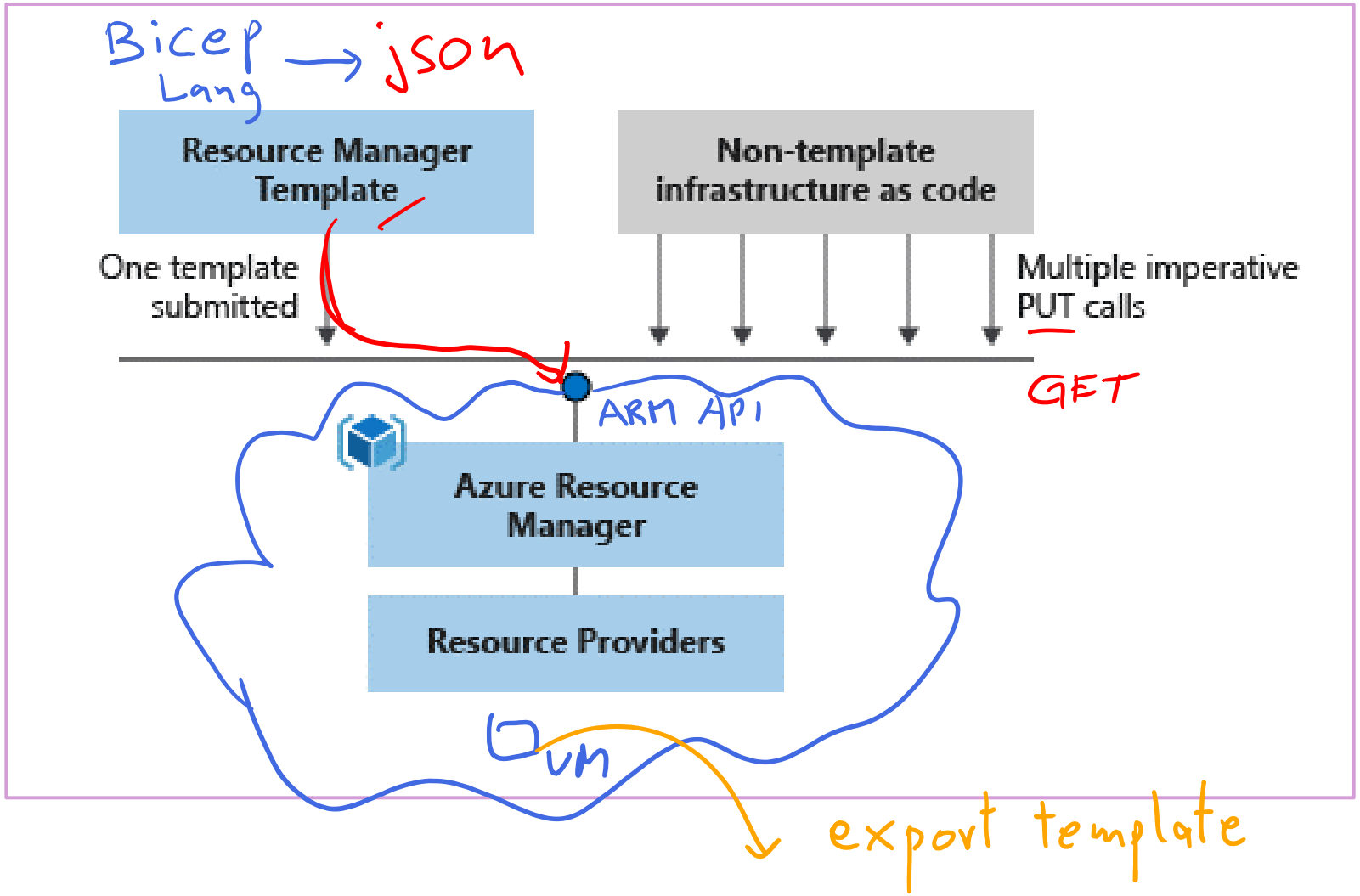
Infra as Code New-AzResourceGroupDeployment - <file> - <RG>

Azure Resource Manager (ARM) templates

Terraform

Azure Resource Manager (ARM) templates are JavaScript Object Notation (JSON) files that can be used to create and deploy Azure infrastructure without having to write programing commands.

- Declarative syntax
- Repeatable results
- Orchestration
- Modular files
- Built-in validation
- Exportable code



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Bicep

export template

Reverse Templating

Bicep

```
param location string = resourceGroup().location
param storageAccountName string = 'toylaunch${uniqueString(resourceGroup().id)}'

resource storageAccount 'Microsoft.Storage/storageAccounts@2021-06-01' = {
  name: storageAccountName
  location: location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  properties: {
    accessTier: 'Hot'
  }
}
```

Symb. Name

RG

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Review the Azure PowerShell module

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Module overview

You can manage Azure resources by using the Azure portal, which is usually the most common administration method. However, for some tasks, PowerShell is more convenient. In this Module, you'll learn about the Azure PowerShell environment and the Az module for Windows PowerShell. Also, you'll learn about ways to manage Microsoft Entra ID by using PowerShell modules.

Units include:

- Azure PowerShell overview
- What is the Azure Az PowerShell module?
- Installing the Azure Az PowerShell module
- Migrate Azure PowerShell from AzureRM to Az
- What are the Microsoft Azure Active Directory Module for Windows PowerShell and Azure Active Directory PowerShell for Graph modules?

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Azure PowerShell overview

- Azure PowerShell is a set of cmdlets for managing Azure resources directly from the PowerShell command line.
- It’s a module that you add to Windows PowerShell or PowerShell Core.
- Azure PowerShell works with PowerShell 5.1 on Windows, PowerShell 7.0.6 LTS, and PowerShell 7.1.3 or newer on all platforms.
- You can use Azure PowerShell in your browser with Azure Cloud Shell, or install it on your local machine on Windows, MacOS, or Linux.

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What is the Azure Az PowerShell module?

- The Az PowerShell module is a set of cmdlets for managing Azure resources directly from PowerShell.
- The Az PowerShell module is the successor of the AzureRM module and is the current version to use for interacting with Azure.
- The Az PowerShell module features:
  - Security and stability.
  - Support for all Azure services.
  - New capabilities.
- You can log issues or feature requests for the Az module directly on the GitHub repository, or through Microsoft support if you have a support contract.

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Installing the Azure Az PowerShell module

- You can install the Azure Az PowerShell module by using one of the following methods:
  - The **Install-Module** cmdlet
  - Azure PowerShell MSI
  - Az PowerShell Docker container
- To install the Az module, run the following command:

  Install-Module -Name Az -Scope CurrentUser -Repository PSGallery –Force
- To sign in to Azure from Azure PowerShell, run the following command:

  Connect-AzAccount

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Migrate Azure PowerShell from AzureRM to Az

- Scripts created for the AzureRM cmdlets won't automatically work with the Az module.
- To make the transition easier, the AzureRM to Az migration toolkit was developed.
- Instead of using **AzureRm** or **Azure** in cmdlet names, you use **Az** cmdlets.
- Automatic migration is the recommended option for migrating from AzureRM to the Az PowerShell module.
- Install the AzureRM to Az migration toolkit by using the following command:

```
Install-Module -Name Az.Tools.Migration
```

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What are the Microsoft Azure Active Directory Module for Windows PowerShell and Azure Active Directory PowerShell for Graph modules?

- You can use the Azure Active Directory Module for Windows PowerShell cmdlets for Microsoft Entra ID administrative tasks such as user management, domain management, and configuring single sign-on.
- Microsoft is replacing the Azure Active Directory Module for Windows PowerShell with Active Directory PowerShell for Graph.
- To install Azure Active Directory Module for Windows PowerShell, use the following command:

  Install-Module MSOnline

- To install Azure Active Directory PowerShell for Graph, use the following command:

  Install-Module AzureAD or Install-module AzureADPreview

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Review the features and tools for Azure Cloud Shell

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Module overview

Instead of using the locally installed PowerShell module for managing Azure resources, you can use the Azure Cloud Shell environment. The Azure Cloud Shell environment enables you to use the PowerShell or Bash environment and commands to manage Azure resources. Azure Cloud Shell is available in the Azure portal and in the Microsoft 365 admin portal. In this Module, you'll learn about Cloud Shell and its features.

Units include:

- Cloud Shell overview
- Features and tools for Azure Cloud Shell
- Demonstration: Use Cloud Shell

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Cloud Shell overview

- Azure Cloud Shell is an interactive, browser-accessible shell for managing Azure resources.
- You can use Cloud Shell to work untethered from a local machine in a way only the cloud can provide.
- You can access the Cloud Shell in three ways:
  - Direct link. Open a browser and refer to **https://shell.azure.com**.
  - Azure portal. Select the Cloud Shell icon on the Azure portal.
  - Code snippets.
- After first launch, you can use the shell type drop-down control to switch between **Bash** and **PowerShell**.

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Features and tools for Azure Cloud Shell

The main features of Azure Cloud Shell are:

- Secure automatic authentication.
- $HOME persistence across sessions.
- Azure drive (Azure:).
- Ability to manage Exchange Online.
- Deep integration with open-source tooling.

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Demonstration: Use Cloud Shell

In this demonstration, you'll learn how to experiment with Azure Cloud Shell.

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Manage Azure resources with Windows PowerShell

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Module overview

Azure virtual machines (VMs) provide a fully configurable and flexible computing environment. You can create and manage these VMs by using the Azure portal, Windows PowerShell with the Az module, or the Cloud Shell environment. In this Module, you'll learn how to create and manage Azure VMs by using PowerShell. Finally, you'll learn about managing Azure storage accounts and Azure subscriptions with PowerShell.

Units:

- Creating Azure VMs with Windows PowerShell
- Managing Azure VMs with Windows PowerShell
- Managing storage with Azure PowerShell
- Managing Azure subscriptions with Azure PowerShell

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Creating Azure VMs with Windows PowerShell

- To create a new Azure VM with PowerShell commands, you can use your locally installed Windows PowerShell with Az module or the Cloud Shell environment available in the Azure portal.
- When using locally installed PowerShell, you first need to use the **Connect-AzAccount** command to authenticate.
- To create and use a new Azure VM, you should:
  1. Create a resource group.
  2. Run the **New-AzVM** command with the configuration parameters.
  3. Find the public IP address.
  4. Connect to the Azure VM.

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Managing Azure VMs with Windows PowerShell

The most common management tasks for Azure VMs are:

- Modifying VM size profile with the **Get-AzVMSize** and **Update-AzVM** commands.
- Starting and stopping the VM with the **Start-AzVM** and **Stop-AzVM** commands.
- Adding or modifying disks to the Azure VM.

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Managing storage with Azure PowerShell

- Azure PowerShell allows you to manage Azure-related storage.
- Before you start managing your storage, you should first create a storage account by using **New-AzStorageAccount**.
- You can use the **Set-AzStorageAccount** cmdlet to modify an Azure Storage account.

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Managing Azure subscriptions with Azure PowerShell

- In Azure PowerShell, accessing the resources for a subscription requires changing the subscription associated with your current Azure session.
- To change subscriptions, you need to first retrieve an Azure PowerShell Context object with **Get-AzSubscription** and then change the current context with **Set-AzContext**.

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Lab – Azure resource management with PowerShell

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Lab: Azure resource management with PowerShell

- **Exercise 1:** Activating the Azure subscription and installing the PowerShell Az module
- **Exercise 2:** Using Azure Cloud Shell
- **Exercise 3:** Managing Azure resources with Azure PowerShell

Sign-in information for the exercises:

Virtual machines:

- **AZ-040T00A-LON-DC1**
- **AZ-040T00A-LON-CL1**

User name: **Adatum\Administrator**

Password: **Pa55w.rd**

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Lab scenario

You're a system administrator for the London branch office of Adatum Corporation. You need to evaluate the Azure platform to run VMs and other resources for your company. As a part of your evaluation, you also want to test PowerShell administration of Azure-based resources.

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Lab-review questions

- Why did you have to use the **Set-ExecutionPolicy** command before installing the Az module for PowerShell?
- What is the purpose of the **Connect-AzAccount** command?
- When you were creating a new VM by using PowerShell, what was the purpose of using the **Get-Credential** command?

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Lab-review answers

- Why did you have to use the **Set-ExecutionPolicy** command before installing the Az module for PowerShell?

  You need to set the PowerShell execution policy to **RemoteSigned** value, so you can pull and install modules from PowerShell Gallery.

- What is the purpose of the **Connect-AzAccount** command?

  You need to run the **Connect-AzAccount** command to authenticate to your Azure tenant that you want to manage.

- When you were creating a new VM by using PowerShell, what was the purpose of using the **Get-Credential** command?

  – You used this command to define local admin credentials for the new Azure VM.

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References

Quickstart: Automatically migrate PowerShell scripts from AzureRM to the Az PowerShell module

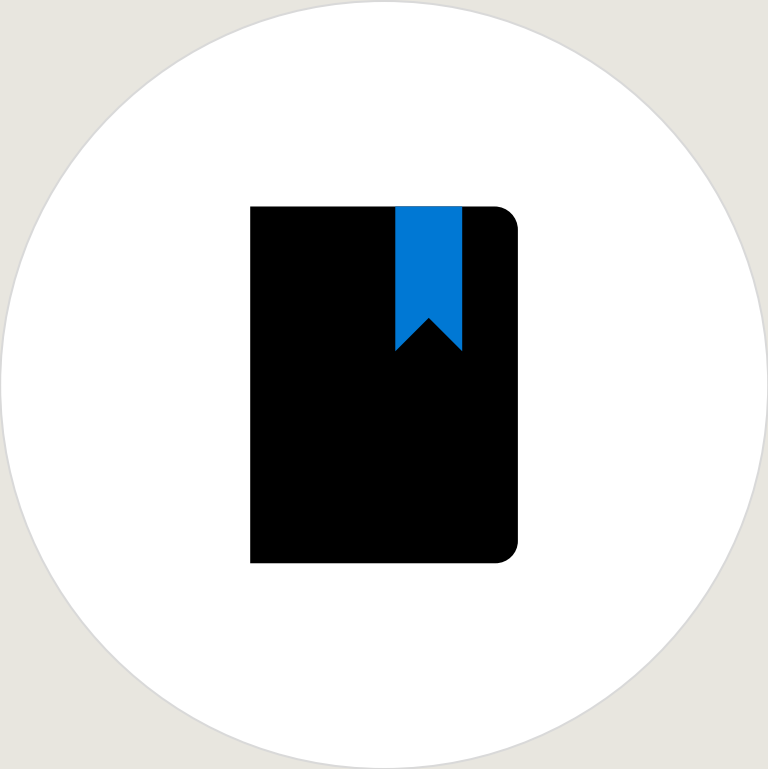
Install Azure Active Directory PowerShell for Graph

AzureAD

MSOnline

New-AzVM

Az.Storage



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Learning Path Recap

In this learning path, we:

- Learned how to install the necessary modules for cloud services management and perform simple administrative tasks on cloud resources.
- Explored the Azure Cloud Shell environment for performing PowerShell-based or Bash-base administration directly from the Azure portal.

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End of presentation

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